

$$\begin{aligned}
 407. \quad \left(4b^2 + \frac{1}{b^2}\right) &= 2 \Rightarrow \left(2b + \frac{1}{b}\right)^2 - 4 = 2 \\
 &\Rightarrow \left(2b + \frac{1}{b}\right)^2 = 6 \Rightarrow \left(2b + \frac{1}{b}\right) = \sqrt{6} \\
 &\Rightarrow \left(2b + \frac{1}{b}\right)^3 = (\sqrt{6})^3 = 6\sqrt{6} \\
 &\Rightarrow 8b^3 + \frac{1}{b^3} + 3 \times 2b \times \frac{1}{b} \cdot \left(2b + \frac{1}{b}\right) = 6\sqrt{6} \\
 &\Rightarrow \left(8b^3 + \frac{1}{b^3}\right) + 6\sqrt{6} = 6\sqrt{6} \\
 &\Rightarrow \left(8b^3 + \frac{1}{b^3}\right) = 0.
 \end{aligned}$$

$$\begin{aligned}
 408. \quad \left(2p + \frac{1}{p}\right) &= 4 \Rightarrow \frac{1}{2} \left(2p + \frac{1}{p}\right) = \frac{1}{2} \times 4 \Rightarrow \left(p + \frac{1}{2p}\right) = 2 \\
 &\Rightarrow \left(p + \frac{1}{2p}\right)^3 = 2^3 = 8 \\
 &\Rightarrow p^3 + \frac{1}{8p^3} + 3p \cdot \frac{1}{2p} \left(p + \frac{1}{2p}\right) = 8 \\
 &\Rightarrow p^3 + \frac{1}{8p^3} + 3 \times \frac{1}{2} \times 2 = 8 \\
 &\Rightarrow p^3 + \frac{1}{8p^3} = 8 - 3 = 5.
 \end{aligned}$$

409. Let the number of cows in the herd be x .
Then, number of legs = $4x$; number of heads = x .
 $\therefore 4x - 2x = 14$ or $2x = 14$ or $x = 7$.

410. Let the number of buffaloes be x and the number of ducks be y .
Then, $4x + 2y = 2(x + y) + 24 \Leftrightarrow 2x = 24 \Leftrightarrow x = 12$.

411. Let the number of hens be x and the number of cows be y .
Then, $x + y = 59$... (i)
and $2x + 4y = 190$ or $x + 2y = 95$... (ii)
Subtracting (i) from (ii), we get: $y = 36$.

412. Let the number of parrots be x and the number of tigers be y .
Then, $x + y = 858$... (i)
and $2x + 4y = 1846$ or $x + 2y = 923$... (ii)
Subtracting (i) from (ii), we get: $y = 65$.
Putting $y = 65$ in (i), we get: $x = 858 - 65 = 793$.

413. Let the number of horses be x . Then, number of men = x .
 $\therefore 4x + \frac{x}{2} \times 2 = 70 \Leftrightarrow 5x = 70 \Leftrightarrow x = 14$.

414. Let the full fare be ₹ x and the reservation charge be ₹ y per ticket.
Then, $x + y = 216$... (i)
and $\frac{3}{2}x + 2y = 327$ or $3x + 4y = 654$... (ii)
Multiplying (i) by 3 and subtracting from (ii), we get: $y = 6$.
Putting $y = 6$ in (i), we get: $x = 210$.

415. Let the distance travelled by taxi be x km.
Then, distance travelled by own car = $(90 - x)$ km.
 $\therefore 7x + 6(90 - x) = 675 \Leftrightarrow x = 675 - 540 = 135$ km.

416. Clearly, first man shakes hand with all other 14 men; second man shakes hand with 13 men (other than first man); third shakes hand with 12 men and so on.
 $\therefore$ Total number of hand-shakes = $(14 + 13 + 12 + \dots + 1)$
 $= \frac{14 \times 15}{2} = 105$.

417. Let there be n children in the group.
Then, first child exchanges gifts with other $(n - 1)$ children; second child exchanges gifts with other $(n - 2)$ children; and so on.
So, number of gifts = $2 [1 + 2 + \dots + (n - 1)]$
 $= 2 \times \frac{n(n - 1)}{2} = n(n - 1)$.
 $\therefore n(n - 1) = 132 \Leftrightarrow n^2 - n - 132 = 0$
 $\Leftrightarrow n^2 - 12n + 11n - 132 = 0$
 $\Leftrightarrow (n - 12)(n + 11) = 0 \Leftrightarrow n = 12$.

418. 1044 candles produce 1044 butts, from which $\frac{1044}{9} = 116$ candles can be formed.

The resultant 116 butts give $\frac{116}{9}$ i.e., 12 candles and 8 butts.
12 resultant butts plus 8 remaining butts = 20 butts from which $\frac{20}{9}$ i.e., 2 candles and 2 butts will be obtained.

In the end, 4 butts will be left behind.
 $\therefore$ Total number of candles formed

$$= (116 + 12 + 2) = 130.$$

419. Number of lines printed in 1 hour = $\frac{176400}{7} = 25200$.
Number of lines printed per minute = $\frac{25200}{60} = 420$.

420. Number of pages typed by 1 man in 1 hour = $\frac{60}{8} = 7.5$.
 $\therefore$ Number of men required = $\frac{1710}{7.5} = 228$.

421. Number of pages read by Ashu from 2: 30 to 3: 20 i.e., in 50 min or $\frac{5}{6}$ hr = $\left(\frac{5}{6} \times 30\right) = 25$.

Let the two read the same page x hours after 3: 20.

Then, pages read by Ashu = $(25 + 30x)$. And, pages read by Neeru = $40x$.

$$\therefore 25 + 30x = 40x \Rightarrow x = \frac{5}{2} = 2\frac{1}{2}.$$

Hence, the two read the same page $2\frac{1}{2}$ hours after 3: 20 i.e., at 5: 50.

$$422. \quad \frac{336}{x} = \frac{192}{400} \Rightarrow x = \frac{336 \times 400}{192} = 700.$$